MASTER'S FINAL PROJECT PROPOSAL

MSc Artificial Intelligence Applied to Sports

Sports Data Campus | UCAM

From Broadcast Video to Tactical Intelligence:
A Computer Vision Pipeline for Set-Piece Analysis in Professional Football

Candidate	Moritz Philipp Haaf
Programme	MSc Artificial Intelligence Applied to Sports
Institution	Sports Data Campus / UCAM
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Project Duration	2 months (May – June 2026)
Supervisor	TBD

1. Description

Modern football analytics has shifted from event-based metrics (xG, xT) towards spatiotemporal models built on tracking data, enabling Pitch Control calculations and action valuation frameworks. However, commercial optical tracking systems from providers such as Opta, StatsBomb or SkillCorner are budget-intensive and cover primarily the top flights of the Big Five leagues. Clubs in second divisions, women's football, youth academies and international scouting contexts are largely excluded from this level of analysis.

This project develops and evaluates a reproducible, open-source Computer Vision pipeline that automatically derives tactical-analytics-relevant metrics directly from monocular broadcast video, with a deliberate and well-motivated focus on set-piece situations (corners and direct free kicks). Set pieces represent approximately 30% of goals scored at international tournament level, are a central tactical preparation task at professional clubs, and provide an analytically favourable Computer Vision context: the camera is near-static, all outfield players are typically visible within a single frame, and the ball position at execution is precisely recorded in open event data, eliminating live ball detection as a confounding variable.

The pipeline implements a five-stage Game State Reconstruction (GSR) workflow, trained and evaluated on the SoccerNet GSR dataset, which provides the broadcast video clips and all required CV annotations (player bounding boxes, team assignments, jersey numbers, camera calibration parameters). SoccerNet GSR is the sole video data source, ensuring full reproducibility and licence compliance:

- Stage 1: Player Detection using YOLOv8, fine-tuned on SoccerNet GSR annotations
- Stage 2: Multi-Object Tracking using ByteTrack with jersey-colour features
- Stage 3: Camera Calibration using keypoint-based sn-calibration (pitch line intersections)
- Stage 4: Team Assignment and Role Classification via HSV jersey histogram clustering
- Stage 5: Coordinate Transformation projecting detections onto a 105 x 68 metre pitch model

On top of this CV pipeline, an analytical layer computes Pitch Control (Spearman 2018 physical model) and VAEP-like action values (socceraction library) for set-piece event sequences derived from SoccerNet clips. The central scientific contribution is the quantitative validation of these broadcast-derived metrics against StatsBomb 360 ground truth from UEFA Euro 2024: the StatsBomb 360 freeze-frame coordinates serve as the reference player-position layer, enabling direct comparison of Pitch Control values computed from pipeline output versus ground-truth spatial data, without any reliance on proprietary or unlicensed broadcast footage.

The project follows the CRISP-DM methodology throughout all six phases (Business Understanding, Data Understanding, Data Preparation, Modelling, Evaluation, Deployment) and all code, notebooks and results will be published as a reproducible public GitHub repository.

2. Objectives

Main Objective

Develop, implement and evaluate a reproducible open-source Computer Vision pipeline that derives Pitch Control and action value metrics from broadcast video for set-piece situations, and validate the pipeline output quantitatively against StatsBomb 360 ground truth from UEFA Euro 2024.

Specific Objectives

- Implement the five GSR pipeline stages using established open-source baselines (YOLOv8, ByteTrack, sn-calibration), optimised for the near-static camera conditions of set-piece sequences.
- Build a set-piece extraction module that filters corner kick and direct free kick sequences from the StatsBomb Euro 2024 event stream and aligns them with corresponding video frames using event-timestamp matching heuristics.
- Integrate an analytical layer computing Pitch Control (Spearman kinematic model, implemented via pitch-aura / Friends of Tracking) and VAEP action values (socceraction library) on set-piece event sequences.

- Validate the pipeline output against StatsBomb 360 Euro 2024 ground truth using Pearson and Spearman correlation for Pitch Control values and MAE/RMSE for VAEP scores across 5 to 10 selected matches.
- Conduct a secondary analysis correlating broadcast-derived Pitch Control at set-piece kick-off moment with the resulting shot xG, as an indirect operational utility test for scouting applications.
- Publish all code, notebooks, results and documentation as a fully reproducible public GitHub repository.

3. Data Sources

All data sources are publicly available under clearly stated licences. No commercial or proprietary data (DFL, Opta, Wyscout) are used. The CV pipeline is trained and evaluated exclusively on SoccerNet GSR clips, which already contain the broadcast video footage needed. StatsBomb Open Data provides the analytical ground truth via 360-degree freeze-frame coordinates, without requiring any separate broadcast video download.

Source	Usage	Licence
SoccerNet GSR Dataset	Primary video source and CV ground truth: broadcast clips with full annotations for player detection, tracking and camera calibration. Training, validation and test splits used throughout.	Academic use
SoccerNet Camera Calibration	Calibration benchmark and evaluation using ProCC protocol and JaC metric (Magera et al., CVPR 2024)	Academic use
StatsBomb Open Data - Euro 2024	Analytical validation ground truth: 51 matches with 360-degree freeze-frame player coordinates per event, used to benchmark Pitch Control and VAEP output against pipeline-derived positions	Non-commercial
StatsBomb Open Data - Leverkusen 23/24	Secondary analytical validation dataset for additional set-piece sequence testing	Non-commercial
SkillCorner Open Data A-League 24/25	Optional: broadcast tracking reference for pipeline output comparison (10 matches)	CC BY-NC-SA
Metrica Sports Sample Data	Optional: synchronised tracking and event data for methodology cross-validation	Research

4. Technologies, Tools and Programming Languages

Primary Language

Python 3.11

Computer Vision and Deep Learning

- Ultralytics YOLOv8 / YOLOv11: player and ball detection backbone, fine-tuned on SoccerNet GSR annotations
- ByteTrack (ECCV 2022): primary multi-object tracker with jersey-colour re-identification features
- sn-calibration / sn-gamestate (SoccerNet): camera calibration baseline and end-to-end GSR reference pipeline
- OpenCV: image processing, homography computation, lens distortion correction
- SAM 2 (Meta AI, optional): segmentation refinement for improved player boundary detection

Football Analytics

- socceraction v1.5.3 (KU Leuven): SPADL action representation and VAEP action value computation
- pitch-aura v0.1.0: Spearman kinematic Pitch Control, Voronoi and tactical metrics
- mplsoccer v1.6.1: pitch visualisation, heatmaps and spatial overlays
- statsbombpy: Python API client for StatsBomb Open Data
- kloppy v3.18.0: vendor-agnostic spatiotemporal data model

Data Engineering and Development

- Pandas, NumPy, PyArrow / Parquet: data manipulation and storage
- scikit-learn: sensitivity analysis and statistical validation
- Poetry: dependency management and reproducible environments
- GitHub Actions: continuous integration and automated testing
- pytest: unit and integration tests on analytical functions
- Jupyter Notebooks: exploratory data analysis and visualisation

Compute and Delivery

- Local GPU (RTX 3060 class) for inference and fine-tuning
- Google Colab Pro: cloud compute backup
- Streamlit: interactive dashboard prototype demonstrating pipeline output on example matches

- YouTube (unlisted): video submission for the explanatory CRISP-DM walkthrough

Methodology

CRISP-DM applied across six explicit phases with defined deliverables per phase, as required by the SDC Final Project guidelines.

5. Expected Value and Outcome

Scientific Contribution

- Quantified Pitch Control accuracy: Pearson and Spearman correlation between broadcast-derived and StatsBomb 360 ground truth Pitch Control values at set-piece moments, reported with explicit confidence intervals and uncertainty bounds.
- Pitch Control to shot quality link: correlation between broadcast-derived spatial control at set-piece kick-off and the resulting shot xG value, providing an indirect operational utility test not yet systematically published in the football analytics literature.
- Error propagation profile: sensitivity analysis mapping per-stage CV errors (detection mAP, tracking IDF1, calibration JaC) to analytical layer accuracy, establishing where investment in CV quality yields the greatest analytical return.
- Documented utility boundaries: explicit statement of the conditions under which broadcast-derived Pitch Control is and is not reliable enough for operational scouting decisions.

Practical Value for Stakeholders

- Football clubs without optical tracking (Championship, 2. Bundesliga, Frauen-Bundesliga, NWSL, second tiers across Europe) gain access to Pitch Control and action value metrics for set-piece preparation, previously available only to top-tier clubs with commercial provider contracts.
- Youth academies and federations can apply quantitative set-piece analysis to competitions and training sessions outside commercial data coverage.
- Scouting departments can apply the pipeline to publicly available match footage to quantify set-piece tactical profiles of opponent teams or scouting targets, reducing reliance on manual video review.

Project Deliverables

Deliverable	Description	Format
Final Project Report (TFM)	3-level structure: Executive Summary, Introduction/Big Picture, Technical Deep Dive following all 11 SDC sections with CRISP-DM methodology	PDF, 25-40 pages
Explanatory Video	5-8 minute walkthrough of the CRISP-DM phases applied in the project, with pipeline demo on a Euro 2024 set-piece clip	YouTube (unlisted)
GitHub Repository	Modular Python pipeline, Jupyter notebooks, tests, CI, README, setup instructions, all open-source	Public GitHub
Streamlit Dashboard	Interactive prototype demonstrating player minimap overlay and Pitch Control heatmap on example set pieces	Streamlit app
Bibliography PDF	All references, datasets and tool citations formatted consistently	PDF

6. Project Timeline (8 Weeks)

The project runs from 1 May to 30 June 2026. Each fortnight has a buffer week to absorb unexpected delays.

Weeks	CRISP-DM Phase	Focus	Key Milestone
1-2	Business Understanding + Data Understanding	Literature consolidation, SoccerNet GSR repo local setup, StatsBomb 360 Euro 2024 data loaded, set-piece extraction module built and validated	sn-gamestate running on one test clip; set-piece filter producing clean event sequences
3-4	Data Preparation + Modelling (Stages 1-3)	YOLOv8 detection fine-tuned on SoccerNet, ByteTrack tracking operational, camera calibration running on set-piece frames (sn-calibration)	Detection mAP and tracking IDF1 measured on SoccerNet test set; calibration JaC computed
5-6	Modelling (Stages 4-5) + Evaluation (CV layer)	Team assignment, coordinate transformation, end-to-end GSR pipeline complete on Euro 2024 set-piece sequences; GS-HOTA measured	Full minimap output for 5+ Euro 2024 set-piece moments; domain generalisation confirmed
7	Evaluation (Analytical layer)	Pitch Control and VAEP computed on pipeline output; Pearson/Spearman correlation and	Quantitative validation results complete; sensitivity analysis drafted

Weeks	CRISP-DM Phase	Focus	Key Milestone
		MAE/RMSE against StatsBomb 360 ground truth; secondary xG correlation analysis	
8	Deployment + Documentation	Streamlit dashboard, explanatory video, final report written, GitHub repository cleaned and published	All deliverables submitted by 30 June 2026

7. Core References

Somers, V. et al. (2024). *SoccerNet Game State Reconstruction: End-to-End Athlete Tracking and Identification on a Minimap.* CVPR Workshops (CVsports). arXiv:2404.11335

Giancola, S. et al. (2025). *SoccerNet 2025 Challenges Results.* arXiv:2508.19182

Magera, F. et al. (2024). *A Universal Protocol to Benchmark Camera Calibration for Sports (ProCC).* CVPR Workshops

Magera, F. et al. (2025). *BroadTrack: Broadcast Camera Tracking for Soccer.* WACV 2025. arXiv:2412.01721

Theiner, J. & Ewerth, R. (2023). *TVCalib: Camera Calibration for Sports Field Registration.* WACV 2023. arXiv:2207.11709

Spearman, W. (2018). *Beyond Expected Goals.* MIT Sloan Sports Analytics Conference

Decroos, T. et al. (2019). *Actions Speak Louder Than Goals: Valuing Player Actions in Soccer (VAEP).* SIGKDD 2019. arXiv:1802.07127

Fernandez, J., Bornn, L. & Cervone, D. (2021). *A Framework for the Fine-Grained Evaluation of the Instantaneous Expected Value of Soccer Possessions (EPV).* Machine Learning, 110

Zhang, Y. et al. (2022). *ByteTrack: Multi-Object Tracking by Associating Every Detection Box.* ECCV 2022. arXiv:2110.06864

Bassek, M. et al. (2025). *Integrated Dataset of Spatiotemporal and Event Data in Elite Soccer.* Scientific Data. DOI: 10.1038/s41597-025-04505-y

Van Haaren, J. (2025). *Soccer Analytics 2025 Review.* janvanhaaren.be